Supplementary Material B

Implementation details and additional results

to *An Adjusted Ersatz Discrepancy for Global Sensitivity Analysis: Theory and Benchmarks*

B.1 Implementation

All computations use the Python module `discrepancy.py` (version 2.0), which implements the estimators, the test functions and the designs, with NumPy, SciPy and, for the gradient-boosting surrogate of HYMOD only, scikit-learn. The scripts of the repository produce the results as follows:

- `reference_indices.py`, `delta_reference.py` and `hymod_reference.py`: reference values of T_i and δ_i (Table S1);
- `experiments.py`: benchmark, convergence, Becker and HYMOD experiments, and the sensitivity analysis of the method;
- `null_distribution.py`: distribution of the adjusted ersatz under independence (Supplementary Material A, Table 2);
- `theory_checks.py`: numerical steps of Supplementary Material A;
- `summaries.py`, `make_tables.py` and `make_figures.py`: tables and figures.

Every random stream is seeded, so all tables and figures are reproducible.

Designs. Scrambled Sobol' points are the first N points of `scipy.stats.qmc.Sobol` with its default scrambling, a linear matrix scramble followed by a digital shift. Maximin Latin hypercubes are the best of 20 random Latin hypercubes (`scipy.stats.qmc.LatinHypercube`) by minimum distance between points. Simple random samples are independent uniform points.

Ersatz statistics. Algorithm 1 of the main text, with $s = \lceil N^{\alpha_g} \rceil$, $\alpha_g = 1/2$, $\theta_f = 1/2$, ordinal ranks (ties broken by order of appearance) and a non-cascading fill on in-grid Moore neighbourhoods. The S-ersatz uses the min-max scaled output and no fill.

Polynomial chaos expansion and Shapley effects. Full total-degree bases of orthonormal Legendre polynomials on $[0, 1]$ (Hermite polynomials of the standard normal inputs of the Oakley–O'Hagan function); ordinary least squares through a QR decomposition; degree $p \in \{1, \ldots, 10\}$ selected by the corrected leave-one-out error of Blatman and Sudret (2010) among the degrees with $N \geq 2P$ and $P \leq 1000$ terms. Total-order indices and Shapley effects are computed from the squared coefficients of the orthonormal basis (main text, Section 2.6).

PAWN. Given-data version with $n = 10$ equal-width conditioning intervals on the probability scale and the maximum of the two-sample Kolmogorov–Smirnov statistics (`scipy.stats.ks_2samp`) between the full output sample and each conditional sub-sample.

Becker metafunction. Port of `sensobol::metafunction` with $k_2 = 0.5$ and $k_3 = 0.2$: thirteen univariate functions (linear, quadratic, cubic, exponential, periodic, discontinuous, non-monotonic, inverse, no effect, trigonometric, large and small piecewise, oscillation), coefficients drawn from $0.7\,N(0, 0.5) + 0.3\,N(0, 5)$, $\lceil k/2 \rceil$ pairwise and $\lceil k/5 \rceil$ three-way interactions (one if $k = 3$). Interaction terms are evaluated row-wise for every k; `sensobol` collapses a single sampled pair or triplet to a constant, which we do not reproduce. Input distributions: uniform, normal $N(0.5, 0.15)$, Beta$(8, 2)$, Beta$(2, 8)$, Beta$(2, 0.8)$, Beta$(0.8, 2)$ and logit-normal$(0, 3.16)$, all truncated to $[0, 1]$, or one of them drawn per input.

References. T_i: analytical, or Jansen estimates on scrambled Sobol' designs (base size 2^{17} for Bratley et al. 1992, 2^{16} for Oakley–O'Hagan and Multiverse, 2^{13} for each Becker configuration); for HYMOD, Jansen estimates (2^{15}) on a degree-7 PCE and on a gradient-boosting surrogate of the 50,000-run sample, which agree to within 0.002 (the table reports their average). δ_i: plug-in histogram estimates on 64×64 equiprobable cells with $N = 2^{24}$ points (HYMOD: 2^{23} points of the PCE surrogate). Table S1 lists the values.

Test case	Reference	Values (inputs in order)
Bratley–Fox (1988)	T_i	0.278, 0.278, 0.278, 0.278, 0.278, 0.278, 0.278, 0.278
	δ_i	0.117, 0.117, 0.117, 0.117, 0.117, 0.117, 0.117, 0.117
Bratley et al. (1992)	T_i	0.747, 0.254, 0.082, 0.029, 0.008, 0.004, 0.001, 0.001
	δ_i	0.536, 0.178, 0.083, 0.040, 0.019, 0.012, 0.007, 0.007
Ishigami	T_i	0.558, 0.442, 0.244
	δ_i	0.256, 0.422, 0.201
Oakley–O'Hagan	T_i	0.059, 0.063, 0.036, 0.055, 0.024, 0.041, 0.058, 0.082, 0.097, 0.036, 0.151, 0.148, 0.142, 0.141, 0.155
	δ_i	0.030, 0.017, 0.030, 0.025, 0.019, 0.056, 0.056, 0.064, 0.068, 0.044, 0.119, 0.134, 0.119, 0.116, 0.129
Sobol' G	T_i	0.787, 0.242, 0.034, 0.010, 0.000, 0.000, 0.000, 0.000
	δ_i	0.543, 0.189, 0.051, 0.028, 0.007, 0.007, 0.007, 0.007
Multiverse	T_i	0.130, 0.130, 0.130, 0.685, 0.113
	δ_i	0.136, 0.136, 0.136, 0.432, 0.194
HYMOD	T_i	0.241, 0.025, 0.403, 0.004, 0.704
	δ_i	0.215, 0.100, 0.159, 0.056, 0.332

Table S1: Reference total-order indices T_i and moment-independent measures δ_i. The inputs of the Bratley–Fox function, and the three continuous inputs of the Multiverse function, have equal values by symmetry. The δ_i of the four non-influential inputs of the Sobol' G function (0.007) are the residual bias of the reference estimator.

B.2 Per-input estimates at $N = 512$

Tables S2–S8 report, for every input, the median and the inter-quartile range (in brackets) of each estimator over 100 replicates of the scrambled Sobol' design ($N = 512$), and the full-sample values for HYMOD.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
x_1	0.278	0.117	0.849 [0.84,0.87]	0.091 [0.08,0.10]	0.139 [0.10,0.17]	0.110 [0.09,0.14]	0.298 [0.28,0.31]
x_2	0.278	0.117	0.850 [0.83,0.86]	0.095 [0.08,0.10]	0.128 [0.09,0.16]	0.109 [0.08,0.14]	0.296 [0.28,0.32]
x_3	0.278	0.117	0.849 [0.84,0.87]	0.092 [0.08,0.10]	0.160 [0.12,0.19]	0.133 [0.10,0.15]	0.298 [0.28,0.31]
x_4	0.278	0.117	0.849 [0.83,0.87]	0.091 [0.08,0.10]	0.143 [0.12,0.18]	0.121 [0.10,0.15]	0.302 [0.29,0.32]
x_5	0.278	0.117	0.849 [0.84,0.87]	0.096 [0.09,0.11]	0.146 [0.10,0.18]	0.117 [0.08,0.15]	0.298 [0.28,0.32]
x_6	0.278	0.117	0.849 [0.84,0.87]	0.091 [0.08,0.10]	0.164 [0.12,0.19]	0.131 [0.09,0.15]	0.294 [0.28,0.32]
x_7	0.278	0.117	0.847 [0.83,0.86]	0.093 [0.08,0.10]	0.138 [0.09,0.18]	0.118 [0.08,0.15]	0.300 [0.29,0.32]
x_8	0.278	0.117	0.851 [0.84,0.87]	0.093 [0.08,0.11]	0.140 [0.10,0.19]	0.123 [0.09,0.16]	0.302 [0.29,0.32]

Table S2: Bratley–Fox (1988) product function: reference values and estimates.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
x_1	0.747	0.536	0.618 [0.61,0.62]	0.474 [0.47,0.48]	0.747 [0.75,0.75]	0.704 [0.70,0.71]	0.827 [0.82,0.84]
x_2	0.254	0.178	0.414 [0.41,0.42]	0.083 [0.08,0.09]	0.253 [0.25,0.25]	0.210 [0.21,0.21]	0.336 [0.32,0.35]
x_3	0.082	0.083	0.410 [0.40,0.42]	0.025 [0.02,0.03]	0.081 [0.08,0.08]	0.059 [0.06,0.06]	0.188 [0.18,0.20]
x_4	0.029	0.040	0.415 [0.41,0.42]	0.026 [0.02,0.03]	0.028 [0.03,0.03]	0.019 [0.02,0.02]	0.143 [0.13,0.16]
x_5	0.008	0.019	0.414 [0.41,0.42]	0.023 [0.02,0.03]	0.008 [0.01,0.01]	0.005 [0.00,0.00]	0.116 [0.11,0.13]
x_6	0.004	0.012	0.408 [0.40,0.41]	0.020 [0.01,0.03]	0.003 [0.00,0.00]	0.002 [0.00,0.00]	0.098 [0.09,0.11]
x_7	0.001	0.007	0.406 [0.40,0.42]	0.021 [0.01,0.03]	0.001 [0.00,0.00]	0.000 [0.00,0.00]	0.097 [0.09,0.10]
x_8	0.001	0.007	0.408 [0.40,0.42]	0.028 [0.02,0.04]	0.001 [0.00,0.00]	0.000 [0.00,0.00]	0.108 [0.10,0.12]

Table S3: Bratley et al. (1992) function: reference values and estimates.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
x_1	0.558	0.256	0.596 [0.58,0.61]	0.223 [0.21,0.24]	0.558 [0.56,0.56]	0.436 [0.44,0.44]	0.337 [0.32,0.35]
x_2	0.442	0.422	0.618 [0.61,0.63]	0.317 [0.30,0.33]	0.442 [0.44,0.44]	0.442 [0.44,0.44]	0.518 [0.51,0.53]
x_3	0.244	0.201	0.561 [0.55,0.57]	0.140 [0.12,0.16]	0.244 [0.24,0.24]	0.122 [0.12,0.12]	0.221 [0.21,0.24]

Table S4: Ishigami function: reference values and estimates.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
x_1	0.059	0.030	0.509 [0.49,0.53]	0.043 [0.04,0.05]	0.061 [0.06,0.06]	0.031 [0.03,0.03]	0.154 [0.14,0.17]
x_2	0.063	0.017	0.510 [0.49,0.52]	0.040 [0.03,0.05]	0.065 [0.06,0.07]	0.033 [0.03,0.03]	0.143 [0.13,0.16]
x_3	0.036	0.030	0.513 [0.50,0.53]	0.049 [0.04,0.06]	0.037 [0.03,0.04]	0.019 [0.02,0.02]	0.161 [0.15,0.18]
x_4	0.055	0.025	0.509 [0.49,0.53]	0.040 [0.03,0.05]	0.057 [0.05,0.06]	0.030 [0.03,0.03]	0.144 [0.13,0.16]
x_5	0.024	0.019	0.512 [0.49,0.53]	0.043 [0.03,0.05]	0.025 [0.02,0.03]	0.014 [0.01,0.02]	0.153 [0.14,0.17]
x_6	0.041	0.056	0.514 [0.50,0.53]	0.056 [0.05,0.06]	0.043 [0.04,0.05]	0.033 [0.03,0.03]	0.203 [0.19,0.23]
x_7	0.058	0.056	0.516 [0.50,0.53]	0.053 [0.05,0.06]	0.060 [0.06,0.06]	0.042 [0.04,0.04]	0.218 [0.19,0.24]
x_8	0.082	0.064	0.512 [0.50,0.53]	0.057 [0.05,0.06]	0.085 [0.08,0.09]	0.055 [0.05,0.06]	0.203 [0.18,0.22]
x_9	0.097	0.068	0.514 [0.50,0.53]	0.056 [0.05,0.06]	0.101 [0.10,0.10]	0.074 [0.07,0.08]	0.246 [0.22,0.27]
x_{10}	0.036	0.044	0.508 [0.49,0.53]	0.047 [0.04,0.06]	0.037 [0.04,0.04]	0.026 [0.02,0.03]	0.179 [0.16,0.19]
x_{11}	0.151	0.119	0.525 [0.50,0.54]	0.089 [0.08,0.10]	0.155 [0.15,0.16]	0.128 [0.13,0.13]	0.322 [0.30,0.35]
x_{12}	0.148	0.134	0.528 [0.51,0.55]	0.105 [0.09,0.12]	0.150 [0.15,0.15]	0.143 [0.14,0.15]	0.334 [0.31,0.36]
x_{13}	0.142	0.119	0.523 [0.51,0.54]	0.085 [0.08,0.10]	0.140 [0.14,0.14]	0.119 [0.12,0.12]	0.305 [0.29,0.33]
x_{14}	0.141	0.116	0.520 [0.51,0.54]	0.085 [0.07,0.10]	0.140 [0.14,0.14]	0.120 [0.12,0.12]	0.302 [0.27,0.33]
x_{15}	0.155	0.129	0.526 [0.51,0.54]	0.104 [0.09,0.11]	0.149 [0.14,0.15]	0.132 [0.13,0.14]	0.321 [0.30,0.34]

Table S5: Oakley–O'Hagan function: reference values and estimates.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
x_1	0.787	0.543	0.650 [0.64,0.66]	0.495 [0.49,0.50]	0.769 [0.77,0.77]	0.769 [0.77,0.77]	0.725 [0.71,0.73]
x_2	0.242	0.189	0.469 [0.46,0.48]	0.121 [0.11,0.13]	0.192 [0.19,0.19]	0.192 [0.19,0.19]	0.339 [0.33,0.35]
x_3	0.034	0.051	0.449 [0.43,0.46]	0.036 [0.03,0.05]	0.027 [0.03,0.03]	0.026 [0.03,0.03]	0.152 [0.14,0.16]
x_4	0.010	0.028	0.451 [0.44,0.46]	0.031 [0.02,0.05]	0.009 [0.01,0.01]	0.008 [0.01,0.01]	0.127 [0.11,0.14]
x_5	0.000	0.007	0.446 [0.43,0.46]	0.023 [0.02,0.03]	0.001 [0.00,0.00]	0.001 [0.00,0.00]	0.101 [0.09,0.11]
x_6	0.000	0.007	0.448 [0.44,0.46]	0.025 [0.02,0.04]	0.001 [0.00,0.00]	0.001 [0.00,0.00]	0.109 [0.10,0.12]
x_7	0.000	0.007	0.446 [0.43,0.46]	0.029 [0.02,0.05]	0.001 [0.00,0.00]	0.001 [0.00,0.00]	0.117 [0.10,0.13]
x_8	0.000	0.007	0.454 [0.44,0.47]	0.040 [0.03,0.07]	0.002 [0.00,0.00]	0.001 [0.00,0.00]	0.117 [0.10,0.14]

Table S6: Sobol' G function: reference values and estimates.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
θ_1	0.130	0.136	0.493 [0.49,0.50]	0.060 [0.05,0.07]	0.117 [0.11,0.12]	0.103 [0.10,0.11]	0.251 [0.24,0.26]
θ_2	0.130	0.136	0.495 [0.49,0.50]	0.062 [0.05,0.07]	0.121 [0.12,0.12]	0.105 [0.10,0.11]	0.261 [0.25,0.27]
θ_3	0.130	0.136	0.497 [0.49,0.50]	0.064 [0.06,0.07]	0.124 [0.12,0.13]	0.105 [0.10,0.11]	0.256 [0.24,0.27]
ξ	0.685	0.432	0.619 [0.61,0.63]	0.355 [0.35,0.36]	0.650 [0.64,0.65]	0.642 [0.64,0.65]	0.759 [0.75,0.77]
ζ	0.113	0.194	0.507 [0.50,0.51]	0.089 [0.08,0.10]	0.085 [0.08,0.09]	0.047 [0.04,0.05]	0.170 [0.16,0.19]

Table S7: Multiverse function: reference values and estimates.

Input	T_i	δ_i	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
Cmax	0.241	0.215	0.821	0.181	0.241	0.180	0.530
bexp	0.025	0.100	0.816	0.095	0.025	0.020	0.181
alpha	0.403	0.159	0.826	0.128	0.403	0.255	0.219
Rs	0.004	0.056	0.808	0.049	0.004	0.002	0.100
Rq	0.704	0.332	0.856	0.299	0.703	0.543	0.528

Table S8: HYMOD (full sample of 50,000 runs): reference values and estimates.

B.3 Other designs

Tables S9 and S10 repeat Table 2 of the main text for maximin Latin hypercube and simple random sampling. HYMOD is not included for the Latin hypercube design because its sample is fixed; its SRS rows use random subsets of the 50,000 runs.

Test case	Metric	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
Bratley–Fox (1988)	ρ_T	–	–	–	–	–
	MAE_T	0.579	0.178	0.125	0.153	**0.043**
	ρ_δ	–	–	–	–	–
	MAE_δ	0.740	**0.021**	0.053	0.034	0.199
Bratley et al. (1992)	ρ_T	0.86	0.91	**1.00**	**1.00**	0.93
	MAE_T	0.344	0.073	**0.001**	0.016	0.149
	ρ_δ	0.85	0.91	**1.00**	**1.00**	0.93
	MAE_δ	0.346	**0.033**	0.042	0.035	0.180
Ishigami	ρ_T	0.14	0.14	**1.00**	0.14	0.14
	MAE_T	0.183	0.175	**0.001**	0.082	0.100
	ρ_δ	**1.00**	**1.00**	0.14	**1.00**	**1.00**
	MAE_δ	0.305	**0.054**	0.122	0.093	0.099
Oakley–O'Hagan	ρ_T	0.54	0.68	**0.91**	0.84	0.69
	MAE_T	0.428	0.028	**0.005**	0.019	0.156
	ρ_δ	0.61	0.74	0.79	**0.95**	0.74
	MAE_δ	0.446	0.021	0.021	**0.010**	0.173
Sobol' G	ρ_T	0.86	0.91	**0.99**	**0.99**	0.91
	MAE_T	0.387	0.075	**0.012**	**0.011**	0.151
	ρ_δ	0.84	0.89	**0.97**	**0.97**	0.90
	MAE_δ	0.384	**0.032**	0.037	0.037	0.165
Multiverse	ρ_T	0.79	0.67	**0.93**	**0.93**	**0.93**
	MAE_T	0.322	0.086	**0.020**	0.037	0.135
	ρ_δ	0.82	**0.97**	0.64	0.64	0.64
	MAE_δ	0.330	**0.052**	0.075	0.089	0.166
Becker ensemble	ρ_T	0.87	0.92	**0.98**	**0.98**	0.93
	MAE_T	0.401	0.070	**0.012**	0.013	0.174

Table S9: Agreement with T_i and δ_i at $N = 512$, maximin Latin hypercube design (medians over 100 replicates).

Test case	Metric	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
Bratley–Fox (1988)	ρ_T	–	–	–	–	–
	MAE_T	0.579	0.171	0.128	0.153	**0.043**
	ρ_δ	–	–	–	–	–
	MAE_δ	0.739	**0.018**	0.053	0.036	0.198
Bratley et al. (1992)	ρ_T	0.90	0.91	**1.00**	**1.00**	0.93
	MAE_T	0.350	0.078	**0.001**	0.016	0.152
	ρ_δ	0.90	0.91	**1.00**	**1.00**	0.93
	MAE_δ	0.353	0.036	0.042	**0.035**	0.183
Ishigami	ρ_T	0.14	0.14	**1.00**	0.14	0.14
	MAE_T	0.191	0.169	**0.001**	0.081	0.106
	ρ_δ	**1.00**	**1.00**	0.14	**1.00**	**1.00**
	MAE_δ	0.313	**0.050**	0.122	0.093	0.094
Oakley–O'Hagan	ρ_T	0.56	0.69	**0.88**	0.82	0.69
	MAE_T	0.433	0.026	**0.005**	0.019	0.159
	ρ_δ	0.62	0.74	0.77	**0.95**	0.75
	MAE_δ	0.450	0.023	0.021	**0.010**	0.176
Sobol' G	ρ_T	0.84	0.91	**0.99**	**0.99**	0.89
	MAE_T	0.389	0.081	**0.012**	**0.012**	0.151
	ρ_δ	0.83	0.91	**0.97**	**0.97**	0.88
	MAE_δ	0.388	0.037	**0.036**	0.036	0.168
Multiverse	ρ_T	0.79	0.67	**0.93**	**0.93**	**0.93**
	MAE_T	0.326	0.082	**0.020**	0.037	0.136
	ρ_δ	0.81	**0.97**	0.64	0.64	0.64
	MAE_δ	0.336	**0.044**	0.072	0.088	0.170
HYMOD	ρ_T	0.91	0.91	**1.00**	**1.00**	0.81
	MAE_T	0.475	0.173	**0.005**	0.075	0.199
	ρ_δ	0.96	**1.00**	0.91	0.91	0.96
	MAE_δ	0.578	**0.025**	0.151	0.096	0.202
Becker ensemble	ρ_T	0.87	0.91	**0.98**	**0.98**	0.91
	MAE_T	0.404	0.075	**0.013**	0.013	0.175

Table S10: Agreement with T_i and δ_i at $N = 512$, simple random sampling (medians over 100 replicates; HYMOD: 100 random subsets).

B.4 Screening errors

Test case	θ	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
Bratley–Fox (1988)	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.10	–/0.00	–/0.62	–/0.12	–/0.25	–/0.00
Bratley et al. (1992)	0.01	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.05	1.00/0.00	0.00/0.33	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.50	0.00/0.00	0.00/0.00	0.83/0.00
Ishigami	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.10	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
Oakley–O'Hagan	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	1.00/0.00	0.50/0.27	0.00/0.00	0.00/0.36	1.00/0.00
	0.10	1.00/0.00	0.00/0.60	0.10/0.00	0.00/0.00	1.00/0.00
Sobol' G	0.01	1.00/0.00	1.00/0.00	0.00/0.25	0.00/0.25	1.00/0.00
	0.05	1.00/0.00	0.17/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.83/0.00
Multiverse	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.20	–/0.00
	0.10	–/0.00	–/0.80	–/0.20	–/0.40	–/0.00
HYMOD	0.01	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.05	1.00/0.00	0.50/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
Becker ensemble	0.01	1.00/0.00	0.91/0.01	0.07/0.07	0.04/0.09	1.00/0.00
	0.05	1.00/0.00	0.23/0.07	0.02/0.08	0.02/0.10	1.00/0.00
	0.10	1.00/0.00	0.04/0.15	0.02/0.08	0.02/0.09	0.86/0.00

Table S11: Screening errors $\alpha(\theta)/\beta(\theta)$, scrambled Sobol' design ($N = 512$; medians over 100 replicates; Becker ensemble: means over 512 configurations). A dash marks an undefined error (no input on that side of the threshold).

Test case	θ	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
Bratley–Fox (1988)	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.10	–/0.00	–/0.50	–/0.12	–/0.25	–/0.00
Bratley et al. (1992)	0.01	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.05	1.00/0.00	0.40/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
Ishigami	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.10	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
Oakley–O'Hagan	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	1.00/0.00	0.50/0.27	0.00/0.00	0.00/0.36	1.00/0.00
	0.10	1.00/0.00	0.00/0.40	0.00/0.00	0.00/0.00	1.00/0.00
Sobol' G	0.01	1.00/0.00	1.00/0.00	0.00/0.25	0.00/0.25	1.00/0.00
	0.05	1.00/0.00	0.33/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
Multiverse	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.20	–/0.00
	0.10	–/0.00	–/0.40	–/0.20	–/0.40	–/0.00
Becker ensemble	0.01	1.00/0.00	0.98/0.00	0.17/0.07	0.10/0.08	1.00/0.00
	0.05	1.00/0.00	0.48/0.03	0.03/0.08	0.03/0.09	1.00/0.00
	0.10	1.00/0.00	0.08/0.11	0.02/0.08	0.02/0.10	0.98/0.00

Table S12: Screening errors $\alpha(\theta)/\beta(\theta)$, maximin Latin hypercube design.

Test case	θ	S-ersatz	Adj. ersatz	PCE	Shapley	PAWN
Bratley–Fox (1988)	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.10	–/0.00	–/0.38	–/0.12	–/0.38	–/0.00
Bratley et al. (1992)	0.01	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.05	1.00/0.00	0.60/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
Ishigami	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.10	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
Oakley–O'Hagan	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	1.00/0.00	0.75/0.09	0.00/0.00	0.00/0.36	1.00/0.00
	0.10	1.00/0.00	0.00/0.40	0.00/0.00	0.00/0.00	1.00/0.00
Sobol' G	0.01	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.25	1.00/0.00
	0.05	1.00/0.00	0.67/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
Multiverse	0.01	–/0.00	–/0.00	–/0.00	–/0.00	–/0.00
	0.05	–/0.00	–/0.00	–/0.00	–/0.20	–/0.00
	0.10	–/0.00	–/0.20	–/0.20	–/0.40	–/0.00
HYMOD	0.01	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.05	1.00/0.00	1.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
	0.10	1.00/0.00	0.50/0.00	0.00/0.00	0.00/0.00	1.00/0.00
Becker ensemble	0.01	1.00/0.00	0.99/0.00	0.14/0.07	0.09/0.09	1.00/0.00
	0.05	1.00/0.00	0.69/0.01	0.03/0.08	0.02/0.09	1.00/0.00
	0.10	1.00/0.00	0.12/0.08	0.02/0.08	0.02/0.10	0.98/0.00

Table S13: Screening errors $\alpha(\theta)/\beta(\theta)$, simple random sampling.

B.5 Convergence under simple random sampling

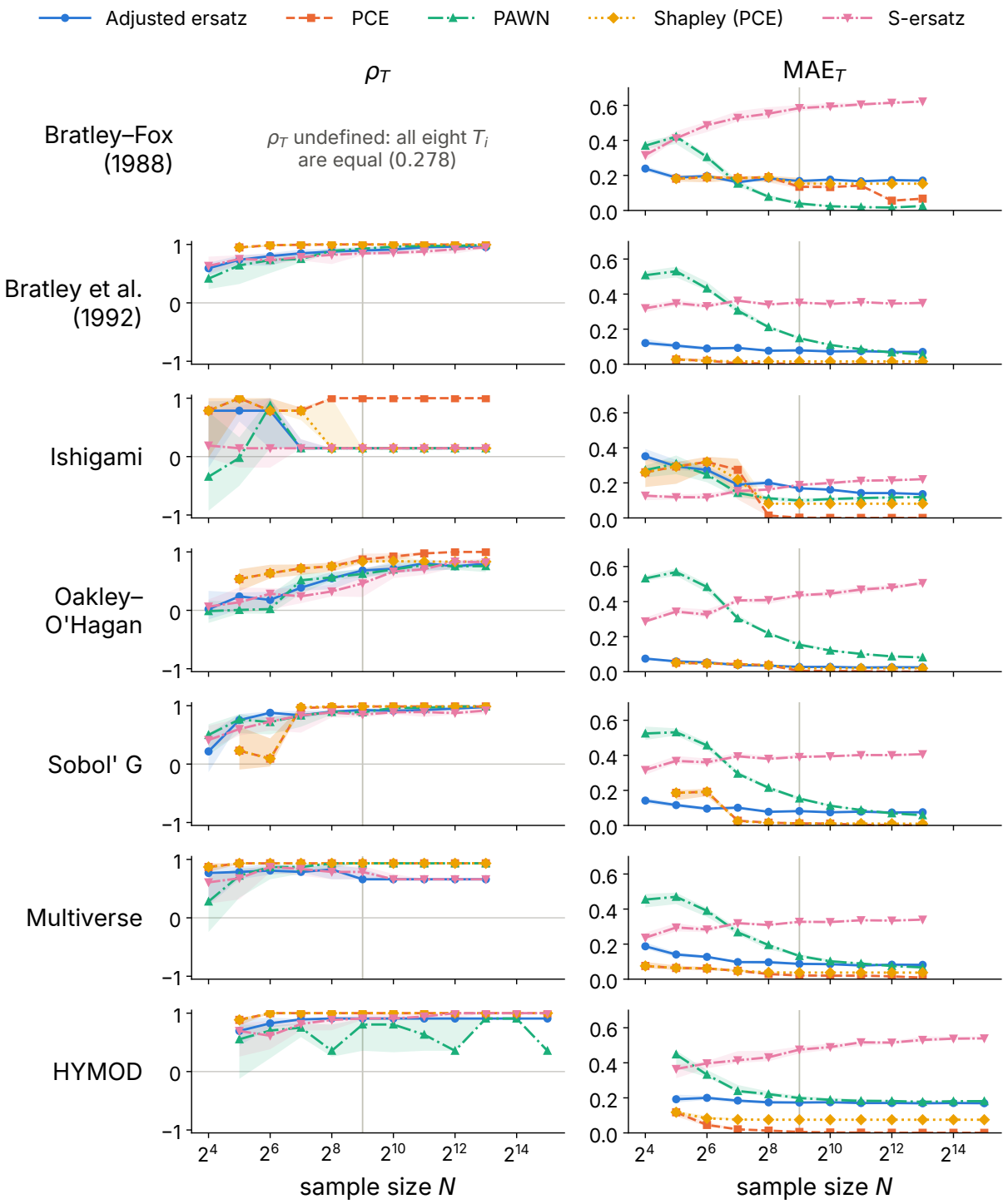


Figure S1: As Figure 6 of the main text, for simple random sampling (HYMOD: random subsets of the 50,000 runs).

B.6 Sensitivity analysis of the method

Parameter	Index	ρ_T	MAE_T	α	β
Adjusted ersatz					
Grid exponent α_g	S_i	0.65 [0.57, 0.74]	0.80 [0.75, 0.86]	0.78 [0.74, 0.81]	0.82 [0.78, 0.86]
	T_i	0.92 [0.86, 0.99]	0.92 [0.88, 0.96]	0.93 [0.90, 0.96]	0.91 [0.87, 0.95]
Fill threshold θ_f	S_i	0.09 [0.06, 0.11]	0.08 [0.06, 0.10]	0.05 [0.04, 0.07]	0.08 [0.07, 0.10]
	T_i	0.35 [0.32, 0.38]	0.20 [0.18, 0.21]	0.20 [0.18, 0.21]	0.17 [0.15, 0.18]
PAWN intervals n		n/a	n/a	n/a	n/a
Screening threshold θ	S_i	n/a	n/a	0.02 [0.01, 0.02]	0.01 [0.01, 0.02]
	T_i	n/a	n/a	0.05 [0.04, 0.05]	0.02 [0.02, 0.02]
Sampling design	S_i	0.00 [0.00, 0.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.01]
	T_i	0.01 [0.01, 0.02]	0.00 [0.00, 0.00]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]
PAWN					
Grid exponent α_g		n/a	n/a	n/a	n/a
Fill threshold θ_f		n/a	n/a	n/a	n/a
PAWN intervals n	S_i	0.42 [0.38, 0.45]	0.86 [0.82, 0.91]	0.27 [0.21, 0.35]	–
	T_i	0.47 [0.45, 0.50]	0.87 [0.84, 0.91]	0.74 [0.66, 0.84]	–
Screening threshold θ	S_i	n/a	n/a	0.16 [0.12, 0.22]	–
	T_i	n/a	n/a	0.63 [0.56, 0.73]	–
Sampling design	S_i	0.53 [0.49, 0.56]	0.12 [0.11, 0.14]	0.04 [0.02, 0.07]	–
	T_i	0.59 [0.56, 0.62]	0.14 [0.13, 0.15]	0.23 [0.20, 0.26]	–

Table S14: First-order (S_i) and total-order (T_i) indices of the five algorithmic parameters with 95% bootstrap intervals (500 resamples of the rows of the design). Scrambled Sobol' design of base size 2^{12} with the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A}_{\mathbf{B}}^{(i)}$, $i = 1, \ldots, 5$ (28,672 evaluations); S_i from the estimator of Saltelli et al. (2010) applied to the centred output, T_i from the estimator of Jansen (1999). The difference $T_i - S_i$ is the contribution of the interactions of the parameter. n/a: the parameter does not enter the metric. –: the type II error of PAWN is zero in 96% of the evaluations (maximum 0.017), so its indices are not reported.

Figure S2 shows how the indices of Table S14 change with the base size N. The first N rows of each matrix of the design of base size 2^{12} form a scrambled Sobol' design of base size N, so the indices for $N = 2^6, \ldots, 2^{12}$ are computed on nested designs without further model runs. Sarrazin et al. (2016) assess the convergence of the ranking, of the screening and of the values of the indices separately, each with a statistic computed from bootstrap resamples. Their ranking statistic, the 95% quantile, over pairs of resamples, of a rank distance weighted by the squared indices, is zero for every metric at $N = 2^{12}$, for S_i as for T_i. For T_i it is below their threshold of one from $N = 2^9$ on, and the ranking by T_i is the same for every metric from $N = 2^7$ on. For the type I error of PAWN the intervals of T_i for n (0.74) and θ (0.63) overlap, but n has the larger T_i in more than 99% of the resamples. The screening statistic, the widest interval among the parameters with $T_i < 0.05$ (the sampling design and the screening threshold on the metrics of the adjusted ersatz), is 0.018 at $N = 2^{12}$, below their threshold of 0.05. The statistic for the values, the widest interval among all parameters, is 0.18. It decreases roughly as $N^{-1/2}$, so a base size of about 2^{16} would be needed to bring it below 0.05. Between $N = 2^{11}$ and $N = 2^{12}$ the largest changes are 0.015 for T_i and 0.023 for S_i. At the present width the second decimal of the indices of the grid exponent is uncertain, but none of the conclusions of Section 3.4 of the main text depends on it.

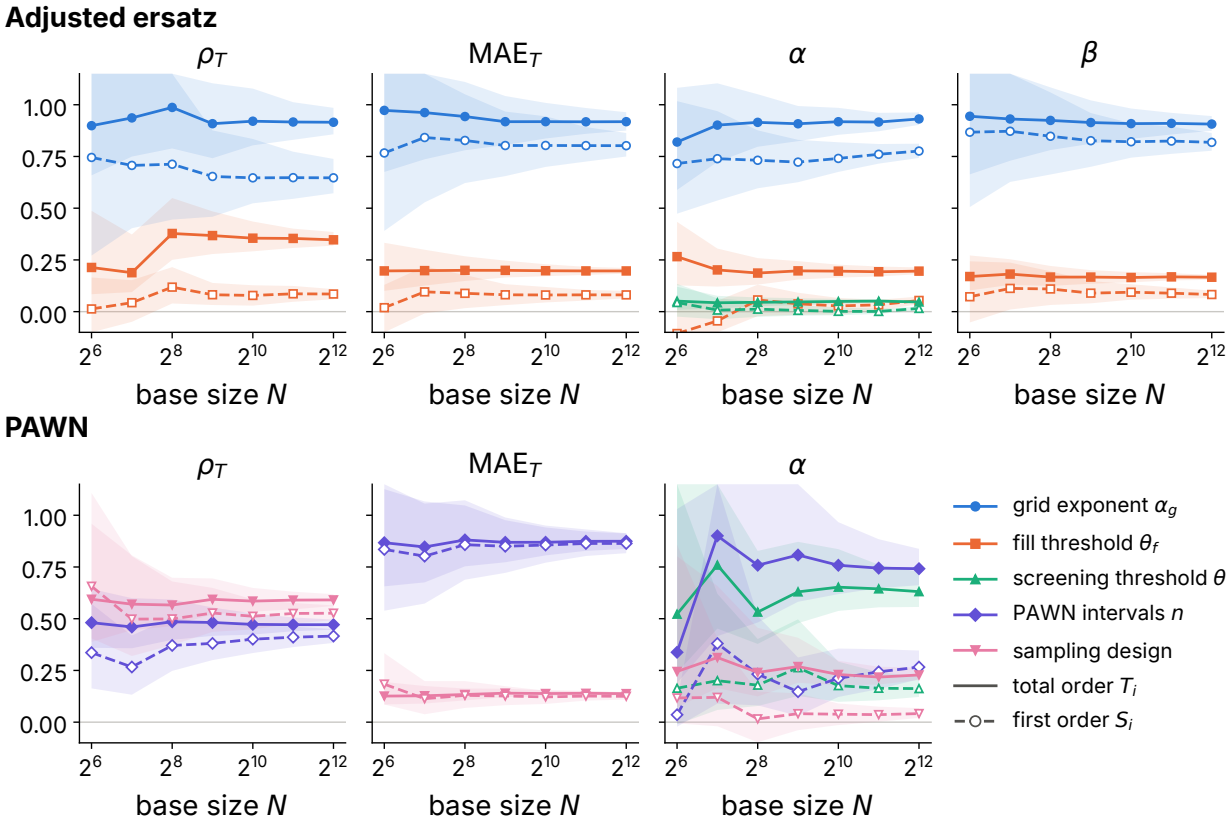


Figure S2: Indices of the algorithmic parameters as functions of the base size N of the nested designs, $N = 2^6, \ldots, 2^{12}$. Solid lines and filled markers: total-order indices T_i; dashed lines and open markers: first-order indices S_i; bands: 95% bootstrap intervals. Parameters with $T_i < 0.03$ at $N = 2^{12}$ (the sampling design for the adjusted ersatz, and the screening threshold for its type II error) are not shown, nor is the type II error of PAWN (Table S14).

B.7 Screening errors and the theory

The script `screening_check.py` of the repository compares the screening errors of the adjusted ersatz with the theory of Supplementary Material A. The theory covers SRS, and under SRS the two agree.

Limits. For the 47 inputs of the six benchmark functions, the limit $\Psi_\lambda(C_i)$ was estimated under SRS at $\lambda = 512/529$, the value of N/s^2 at $N = 512$ ($N = 61{,}952$, $s = 253$), and at $\lambda = 1$ ($N = 65{,}536$, $s = 256$), each with 10 replicates (standard errors at most 0.0011). All 47 limits lie within the bounds $[\delta_i - \varepsilon_\lambda,\ \delta_i + \kappa_\lambda]$ of Supplementary Material A, Proposition 5.4 ($\varepsilon = 0.1208$ and $\kappa = 0.0548$ at $\lambda = 512/529$, $\varepsilon = 0.1400$ and $\kappa = 0.0471$ at $\lambda = 1$). The difference $\Psi_\lambda(C_i) - \delta_i$ ranges from -0.032 to 0.048 at $\lambda = 512/529$ and from -0.039 to 0.041 at $\lambda = 1$. It is largest for inputs almost independent of the output, whose limit is close to κ_λ, and smallest for inputs with δ_i between 0.18 and 0.43. The inputs missed at $\theta = 0.10$ under SRS (Table S13) are those whose limit exceeds the threshold by less than about one standard deviation of the statistic at $N = 512$ (0.017 to 0.022): the eight inputs of the Bratley–Fox function (limit 0.113), x_1 to x_3 of the Multiverse function (0.117), and x_{11}, x_{13} and x_{14} of the Oakley–O'Hagan function (0.108).

Inputs independent of the output. The 345 inputs of the Becker ensemble with $T_i = 0$ are independent of the output (main text, Proposition 3(a)), so under SRS their statistic has the distribution under independence at their sample size, whatever the model (Supplementary Material A, Theorem 4.1). Table S15(a) compares their flag rates with this distribution, simulated with 3000

replicates at each sample size. Under SRS the two agree within one standard error at the three thresholds. The maximin LHS and Sobol' designs give lower values (Supplementary Material A, Section 8 and Table 2).

Sample size. Table S15(b) shows the type I error at $\theta = 0.05$ in the convergence study under SRS, together with the probability that an independent input exceeds 0.05 (Supplementary Material A, Table 2). The type I error oscillates with N/s^2 as this probability does: it is high at $N = 128$, 512 and 2048, where $\kappa_{N/s^2} > 0.05$, and low at $N = 64$, 256, 1024 and 4096, where $\kappa_1 = 0.0471 < 0.05$. For the Bratley et al. (1992) function, whose non-influential inputs have $\delta_i \leq 0.04$, it stays within 0.08 of the probability for an independent input. For the Oakley–O'Hagan function it lies above that probability at every N, by up to 0.37, because the limits of most of its non-influential inputs exceed 0.05. A fixed threshold on the scale of T_i therefore does not give vanishing type I errors as N grows: a consistent test needs a threshold above κ_λ (Supplementary Material A, Corollary 4.5).

Ranking. The 23 pairs of benchmark inputs whose δ_i differ by more than $\varepsilon_\lambda + \kappa_\lambda = 0.1755$ at $\lambda = 512/529$ are ranked as by δ in all 100 replicates at $N = 512$ under SRS and maximin LHS, and in 99.8% of them under the Sobol' design. The 19 pairs whose δ_i differ by more than 0.19, the margin for $\lambda = 1$ quoted in the main text, are ranked as by δ in every replicate of the three designs.

Designs. At $N = 512$ the maximin LHS and Sobol' designs lower the statistic of influential and non-influential inputs alike, relative to SRS: by 0.001 to 0.014 under maximin LHS, and by up to 0.058 under the Sobol' design (median over the inputs of a function from 0.014 to 0.040; only x_8 of the Sobol' G function is not lowered). The Sobol' design lowers it most for the functions dominated by few inputs (Bratley et al. 1992, Multiverse, Sobol' G), as for the first coordinates of the sequence (Supplementary Material A, Section 8). This causes the type II errors of the Bratley et al. (1992) function in Table S11: the means of x_2 and x_3 are 0.083 and 0.026 under the Sobol' design, against 0.141 and 0.077 under SRS. Beyond independence under plain LHS (Supplementary Material A, Proposition 8.1), the theory does not cover these designs.

(a) Becker inputs with $T_i = 0$: share above θ, and mean

Design	$\theta = 0.01$	$\theta = 0.05$	$\theta = 0.10$	Mean
SRS	0.997	0.617	0.026	0.0559
Maximin LHS	0.983	0.359	0.003	0.0466
Scrambled Sobol'	0.942	0.151	0.003	0.0345
Independent input, SRS, same N	0.997	0.608	0.025	0.0562

(b) Type I error at $\theta = 0.05$, SRS, convergence study

N	N/s^2	Bratley et al. (1992)	Oakley–O'Hagan	Sobol' G	Independent input
32	0.889	0.59	0.57	0.51	0.57
64	1.000	0.36	0.36	0.28	0.33
128	0.889	0.79	0.78	0.77	0.72
256	1.000	0.37	0.66	0.43	0.41
512	0.968	0.59	0.75	0.65	0.57
1024	1.000	0.35	0.54	0.49	0.35
2048	0.968	0.77	0.91	0.80	0.71
4096	1.000	0.34	0.64	0.47	0.27
8192	0.989	0.50	0.94	0.59	–

Table S15: Screening errors of the adjusted ersatz and the theory. (a) Inputs of the Becker ensemble with $T_i = 0$ (345 for each design): share of the statistic above θ, and its mean, with the same quantities for an input independent of the output under SRS at the same sample sizes (3000 replicates at each N). Standard errors of the observed shares: at most 0.026. (b) Type I error at $\theta = 0.05$ in the convergence study under SRS (mean over 20 replicates), and probability that an input independent of the output exceeds 0.05 (Supplementary Material A, Table 2; 10,000 replicates).